

Spectrums of substitutional Hamiltonians

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One-dimensional Schrödinger operator on $l^2(\mathbb{Z})$

$$(H\phi)_n = \phi_{n+1} + \phi_{n-1} + v_n\phi_n, \quad n \in \mathbb{Z}, \quad \phi \in l^2(\mathbb{Z})$$

- potential $(v_n)_{n \in \mathbb{Z}}$, bounded real sequence
quasi-periodic: almost Mathieu, Sturm, **substitutional**, ...
- Sturm potential $v_n = \lambda \chi_{[1-\alpha, 1)}(\{n\alpha + \theta\})$ (SturmH $H_{\alpha, \lambda}$).
- operator spectrum
 $\sigma(H) = \{E \in \mathbb{R} : H - EI \text{ has no bounded inverse}\}$

$$H = \begin{pmatrix} \ddots & & \ddots & & \\ & 1 & & v_{-1} & 1 \\ & & 1 & & v_0 & 1 \\ & & & 1 & & v_1 & 1 \\ & & & & \ddots & & \ddots \end{pmatrix}$$

Substitutional potential

- Given a substitutional sequence $w \in \{a, b\}^{\mathbb{Z}}$

$$v_n = f(T^n w), \quad f(\xi) = \begin{cases} \lambda & \xi(0) = a \\ -\lambda & \xi(0) = b \end{cases}, \quad \xi \in \{a, b\}^{\mathbb{Z}}.$$

T is the left shift on $\{a, b\}^{\mathbb{Z}}$, i.e., $T\xi(k) = \xi(k+1)$, $\forall k \in \mathbb{Z}$.

- w_{tm} : generated by Thue-Morse substitution (TMH $H_{1,\lambda}$)

A morphism σ on $\{a, b\}^*$ with $\tau(a) = ab$, $\tau(b) = ba$.

$$\tau(a) = ab, \tau^2(a) = abba, \dots, \tau^\infty(a) = abbabaabbaa \dots$$

$$w_{tm} = \lim_{k \rightarrow \infty} \tau^k(a) \cdot \tau^k(a) = \dots baababba \cdot abbabaab \dots$$

- w_{pd} : period doubling substitution (PDH $H_{pd,\lambda}$)

$$\zeta(a) = ab, \zeta(b) = aa.$$

- w_m : generalized Thue-Morse substitution (GTMH $H_{m,\lambda}$)

$$\eta(a) = a^m b^m, \eta(b) = b^m a^m, \text{ with } m \geq 1$$

- L,Qu,Yao, CMP, 2022. PDH. For $\lambda > 0$,

$$\dim_H \sigma(H_{pd,\lambda}) \geq \frac{\log(1 + \sqrt{5})/2}{\log 4} = 0.34712 \dots$$

We also show an exact coding, integrated density of states, gap labling, and existence of unbounded orbits.

- L,Tang, to appear in AMS, GTMH. For $m \geq 2$, $\lambda > 0$,

$$\dim_H \sigma(H_{m,\lambda}) > \frac{\log \max\{2, m - 3\}}{\log 64m + 4}.$$

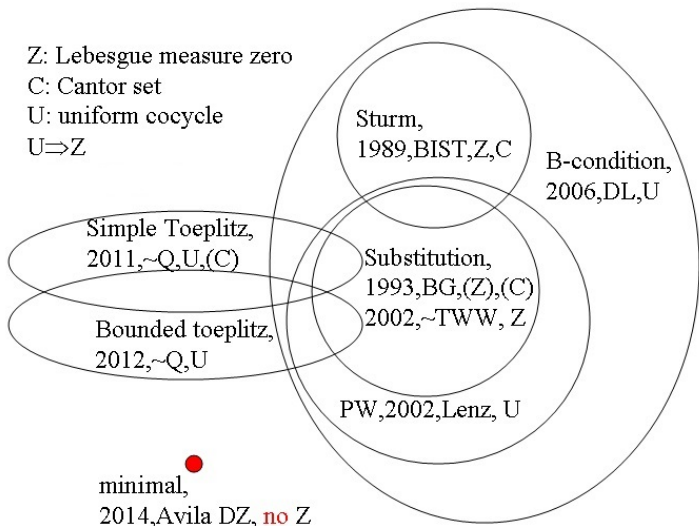
Lebesgue measure zero spectrum

Z: Lebesgue measure zero

C: Cantor set

U: uniform cocycle

$U \Rightarrow Z$



Fractal dimensions of spectrums

- Let $\alpha = [0; a_1, a_2, \dots]$ (SturmH)

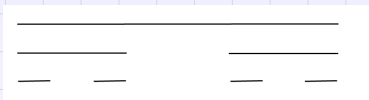
$$K_* = \liminf_n (a_1 \cdots a_n)^{1/n}, \quad K^* = \limsup_n (a_1 \cdots a_n)^{1/n}$$

- L, Wen '04 $\lambda > 20$,
 - if $K_* < \infty$, then $0 < \dim_H \sigma(H_{\lambda, \alpha}) < 1$
 - if $K_* = \infty$, then $\dim_H \sigma(H_{\lambda, \alpha}) = 1$.
- L, Qu, Wen '14 $\lambda > 25$,
 - if $K^* < \infty$, then $0 < \overline{\dim}_B \sigma(H_{\lambda, \alpha}) < 1$
 - if $K^* = \infty$, then $\overline{\dim}_B \sigma(H_{\lambda, \alpha}) = 1$.
- Damanik, Embree, Gorodetski, Tcheremchantsev '07: Fib
 $\lim_{\lambda \rightarrow \infty} (\log \lambda) \dim_H \sigma(H_{\alpha, \lambda}) = \log(1 + \sqrt{2})$
- L, Qu '15, TMH, $\forall \lambda \in \mathbb{R}$, $\dim_H \sigma(H_{1, \lambda}) \geq \frac{\ln 2}{140 \ln 2.1}$.

Hausdorff dimension of classical Cantor set

$$\mathcal{H}_\delta^s(E) = \inf_{\mathcal{C}:\delta\text{-covering}} \sum_{U \in \mathcal{C}} |U|^s, \quad \mathcal{H}^s(E) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s$$

$$\dim_H E = \inf\{s : \mathcal{H}^s(E) = 0\} = \sup\{s : \mathcal{H}^s(E) = \infty\}.$$



$$2^n \left(\frac{1}{3^n}\right)^s = 1 \Leftrightarrow \left(\frac{1}{3}\right)^s + \left(\frac{1}{3}\right)^s = 1 \Leftrightarrow s = \frac{\ln 2}{\ln 3} = \frac{\ln 2^n}{-\ln(1/3)^n}$$

Seperated Nested Structure.

Proposition

Let $\mathcal{J} = \{\mathcal{J}_n : n \geq 0\}$ be a SNS and $A(\mathcal{J}) := \bigcap_{n \geq 0} \bigcup_{J \in \mathcal{J}_n} J$. Assume $\mathcal{J}$ satisfies:

i) There exist $\lambda \in (0, 1)$ and $C > 0$ such that

$$|I| \geq C\lambda^n, \quad \forall n \geq 0, \forall I \in \mathcal{J}_n.$$

ii) There exists $C' > 0$ such that for any $n, k \geq 0$, and $I, I' \in \mathcal{J}_n$,

$$\frac{\#\{J \in \mathcal{J}_{n+k} : J \subset I\}}{\#\{J \in \mathcal{J}_{n+k} : J \subset I'\}} \leq C'.$$

Then the limit set satisfies

$$\dim_H A(\mathcal{J}) \geq \liminf_{n \rightarrow \infty} \frac{\log \#\mathcal{J}_n}{-n \log \lambda}.$$

Transfer matrices

For any $E \in \mathbb{R}$, to solve $Hu = Eu$, we can write

$$u_{n-1} + u_{n+1} + v_n u_n = Eu_n,$$

as

$$\begin{bmatrix} u_{n+1} \\ u_n \end{bmatrix} = \begin{bmatrix} E - v_n & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_n \\ u_{n-1} \end{bmatrix}.$$

Define

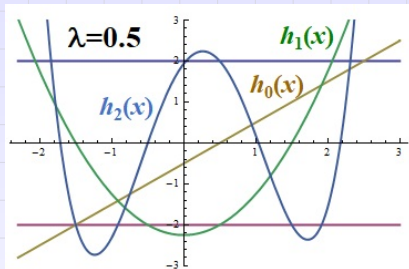
$$T_n(E) = \begin{bmatrix} E - v_n & -1 \\ 1 & 0 \end{bmatrix}$$

We can write

$$\begin{bmatrix} u_{n+1} \\ u_n \end{bmatrix} = T_n(E)T_{n-1}(E) \cdots T_1(E) \begin{bmatrix} u_1 \\ u_0 \end{bmatrix}.$$

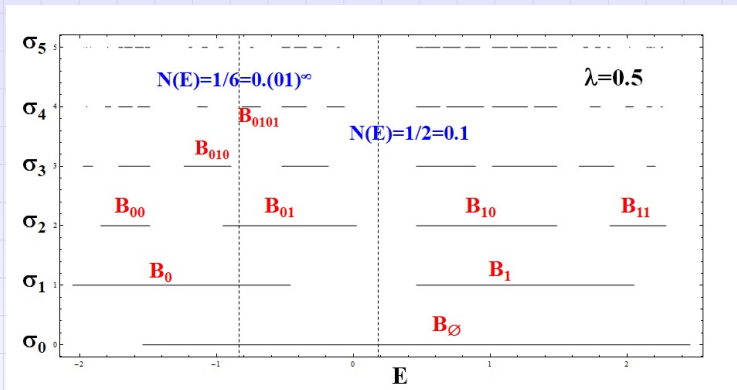
and then define $T_{1 \rightarrow n}(E) = T_n(E)T_{n-1}(E) \cdots T_1(E)$.

Trace polynomial $h_k(x)$ for PDH



- $T_n(x) := \begin{bmatrix} x - v_n & -1 \\ 1 & 0 \end{bmatrix}$, $T_{1 \rightarrow n}(x) := T_n(x) \cdots T_1(x)$.
- $k > 0$, $h_k(x) := \text{tr } T_{1 \rightarrow 2^k}(x)$. Trace polynomials. Degree 2^k .
- $\zeta^\infty(a) = abaaabab \cdots$, $(v_1, v_2, \dots) = (\lambda, -\lambda, \lambda, \lambda, \dots)$.
- $h_0(x) = x - \lambda$, $h_1(x) = x^2 - \lambda^2 - 2$,
- $h_{k+1}(x) = h_k(h_{k-1}^2 - 2) - 2 (\forall k \geq 1)$.
- $M_k := T_{1 \rightarrow 2^k}$, $N_k := T_{2^k+1 \rightarrow 2^{k+1}}$, $M_{k+2} = M_k N_k M_k M_k$
- Cayley-Hamilton Thm $M^2 - (\text{tr } M)M + (\det M)I = 0$.

Periodic approach, Integrated Density of States (IDS)



- $\sigma_k := \{E \in \mathbb{R} : |h_k(E)| \leq 2\}$, 2^k intervals.
- $\sigma(H_{pd,\lambda}) = \bigcap_{k \geq 1} (\sigma_k \cup \sigma_{k+1})$, $(\sigma_k \cup \sigma_{k+1}) \subset (\sigma_{k-1} \cup \sigma_k)$.
- A natural symbol system: $\mathcal{B}_n = \{B_w : w \in \{0,1\}^n\}$.
- $E \in \mathbb{R}$, IDS $N(E) = \lim_{k \rightarrow \infty} \frac{\#\{\text{interval in } \sigma_k \text{ that } < E\}}{\#\{\text{interval in } \sigma_k\}}$.

- Bellissard, Iochum, Scoppola, Testard 1989, $\text{SturmH}(H_{\alpha,\lambda})$

$E \in \sigma(H_{\alpha,\lambda})$ if and only if $(h_k(E))_{k \geq 1}$ is bounded.

- L, Qu, Yao, 2017, $\text{TMH}(H_{1,\lambda})$

There exists uncountable many $E \in \sigma(H_{1,\lambda})$ such that $(h_k(E))_{k \geq 1}$ is unbounded.

- Carvalho, Nonlinearity, 2020, $\text{PDH}(H_{pd,\lambda})$

$E \in \sigma(H_{pd,\lambda})$ if and only if $(h_k(E))_{k \geq 1}$ is bounded.

- L, Qu, Yao, 2022, $\text{PDH}(H_{1,\lambda})$

$\{E \in \sigma(H_{1,\lambda}) \mid (h_k(E))_{k \geq 1} \text{ is unbounded}\}$ is an uncountable dense subset of $\sigma(H_{pd,\lambda})$.

main problem in case of Thue-Morse

Find a Cantor subset in $\{x \in \mathbb{R} : \exists n > 0, \forall k \geq n, 0 \leq h_k(x) \leq 2\}$.

Covering structure:

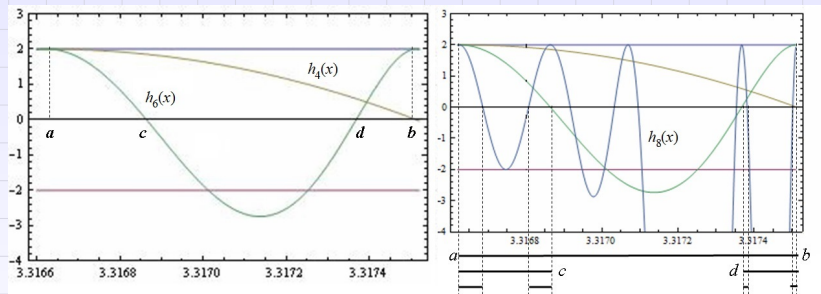


Figure: The graph of $h_4(x)$, $h_6(x)$, $h_8(x)$ for $x \in (3.3166, 3.3175)$.

Problem: How to control length for these bands?

Observation in L. Qu, 2015 for Thue-Morse

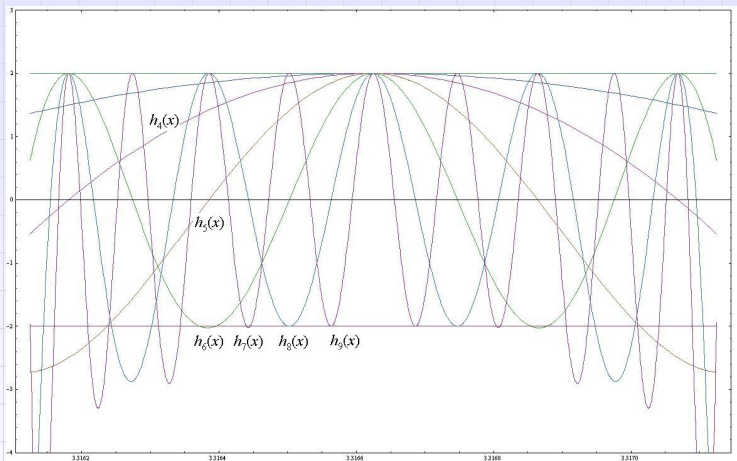


Figure: Graph of $h_3(x), \dots, h_9(x)$, $x \in (3.3161, 3.3172)$, $\lambda = 3$.

Compare with graph of $2 \cos 2^k x$ near 0

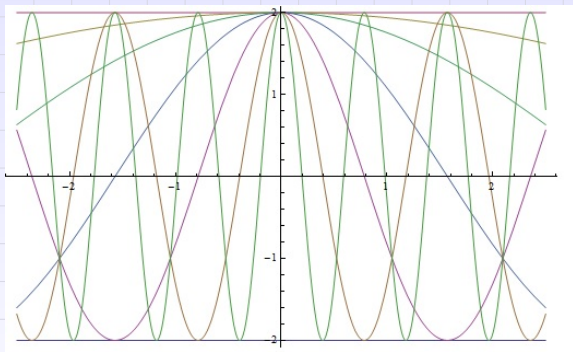
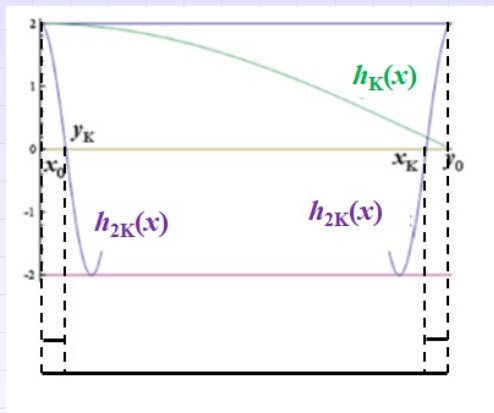


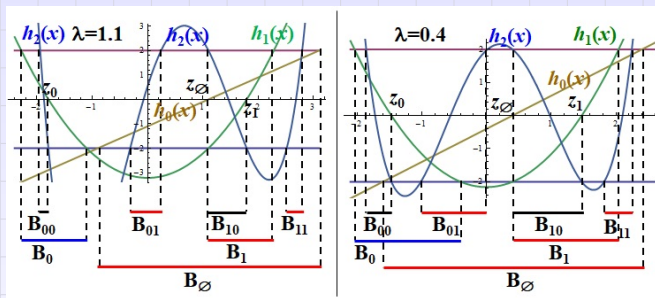
Figure: Graph of $2 \cos 2^k x$, $k = -2, \dots, 3$.



the closeness of h_K to $2 \cos x$ in scale at x_0 , implies the closeness of h_{2K} to $2 \cos x$ in scale at both x_0 and y_0 ,

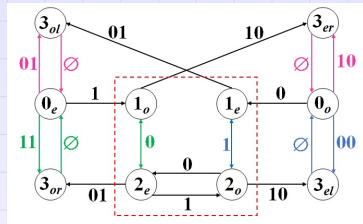
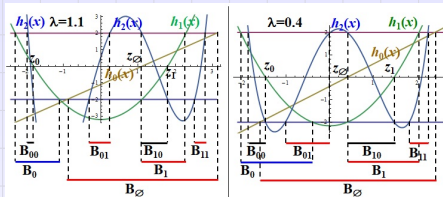
$$\frac{y_0 - x_K}{y_0 - x_0}, \frac{y_K - x_0}{y_0 - x_0} > 2 \cdot 1^{-K}, \dim_H \sigma(H_{1,\lambda}) \geq \frac{\ln 2}{(K \ln 2.1)}$$

coverings of the spectrum



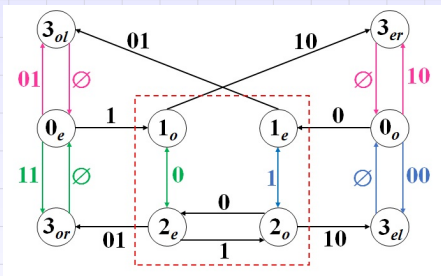
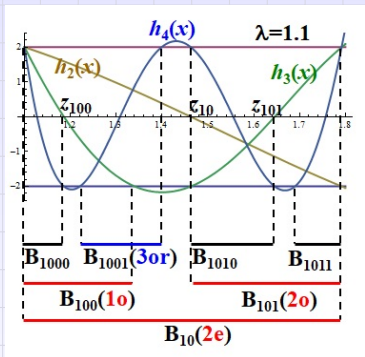
- $\mathcal{B}_n = \{B_w : w \in \{0, 1\}^n\}$
- $\sigma(H_{pd,\lambda}) = \bigcap_{k \geq 1} (\sigma_k \cup \sigma_{k+1}), (\sigma_k \cup \sigma_{k+1}) \subset (\sigma_{k-1} \cup \sigma_k)$
- $\mathcal{B}_n = \{B : B \in \mathcal{B}_n \text{ or } B \in \mathcal{B}_{n+1} \text{ and } B \not\subset \sigma_n\}.$
- $\mathcal{B}_0 = \{B_0, B_\emptyset\}.$
- $\mathcal{B}_1 = \{B_0, B_{01}, B_1, B_{11}\}.$

PDH, another symbolic system, 0_e



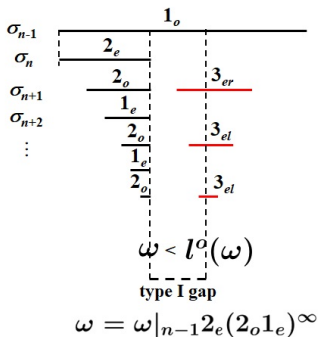
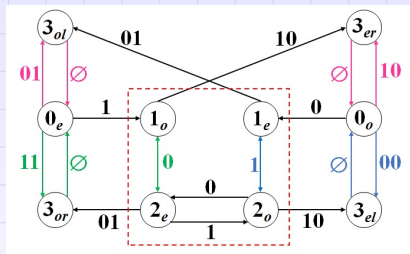
- $\mathcal{A} = \{0_e, 0_o, 1_e, 1_o, 2_e, 2_o, 3_{el}, 3_{er}, 3_{ol}, 3_{or}\}$.
- $h_{k+1}(x) = h_k(h_{k-1}^2 - 2) - 2 (\forall k \geq 1)$.
 $h_k(x) = 0$ implies $h_{k+1}(x) = -2$, $h_n(x) = 2 (\forall n > k + 1)$.
- $\mathcal{B}_n = \{B : B \in \mathcal{B}_n \text{ or } B \in \mathcal{B}_{n+1} \text{ and } B \notin \sigma_n\}$.
- $\mathcal{B}_0 = \{B_0 = I_{3_{el}}, B_\emptyset = I_{0_e}\}$.
- $\mathcal{B}_1 = \{B_0 = I_{3_{el}0_o}, B_{01} = I_{0_e3_{ol}}, B_1 = I_{0_e1_o}, B_{11} = I_{0_e3_{or}}\}$.
- $3_{el} \rightarrow 0_o, 0_e \rightarrow 3_{ol} < 1_o \prec 3_{or}$. Nested Structure.

PDH, nested structure, 2_e



- $h_{k+1}(x) = h_k(h_{k-1}^2 - 2) - 2 (\forall k \geq 1)$.
 $h_k(x) = 0$ implies $h_{k+1}(x) = -2$, $h_n(x) = 2 (\forall n > k + 1)$.
- $2_e \rightarrow 1_o \prec 3_{or} < 2_o$.

Type I gaps

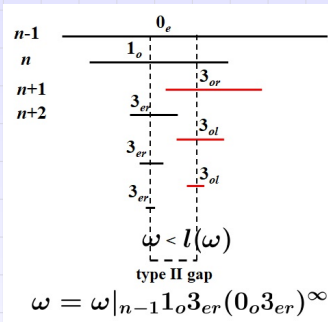
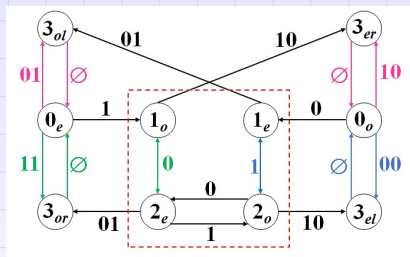


- For $\omega = \omega|_{n-2} 1_o 2_e (2_o 1_e)^\infty$, $l^o(\omega) = \omega|_{n-2} 1_o 3_{er} (0_o 3_{el})^\infty$.
- $\Pi(\omega) = u01^\infty$, $\Pi(l^o(\omega)) = u10^\infty$. $h_k(\pi(\omega)) = 2$ for $k \geq n$.

$$\lim_{k \rightarrow \infty} h_{n+2k}(\pi(l^o(\omega))) = \infty, \quad \lim_{k \rightarrow \infty} h_{n+2k+1}(\pi(l^o(\omega))) = \sqrt{2}$$

- $(h_k(\pi(l^o(\omega))))_{k \geq 0}$ is unbounded.
- for E in the gap, IDS $N(E) = 0.u01^\infty = 0.u10^\infty$.

Type II gaps



- For $\omega = \omega|_{n-2} 0_e 1_o 3_{er} (0_o 3_{er})^\infty$, $l(\omega) = \omega|_{n-2} 0_e 3_{or} (0_e 3_{ol})^\infty$.
- $\Pi(\omega) = \Pi(l(\omega)) = u(10)^\infty \cdot (2_e 2_o)^\infty, (10)^\infty$

$$\lim_{k \rightarrow \infty} h_{2k+n}(\pi(\omega)) = \lim_{k \rightarrow \infty} h_{2k+n+1}(\pi(l(\omega))) = -\infty,$$

$$\lim_{k \rightarrow \infty} h_{2k+n+1}(\pi(\omega)) = \lim_{k \rightarrow \infty} h_{2k+n}(\pi(l(\omega))) = -\sqrt{2}.$$

- for E in the gap, IDS $N(E) = 0.u(10)^\infty$.

Gap labelling and Integrated Density of States

IDS N is constant on a gap G of the spectrum, denoted as $N(G)$.

Theorem

We have the gap labelling:

$$\{N(G) : G \in \mathcal{G}_I\} = \mathcal{D} \cap (0, 1) \quad \text{and} \quad \{N(G) : G \in \mathcal{G}_{II}\} = \mathcal{E},$$

where $\mathcal{D}$ is the set of positive dyadic numbers and

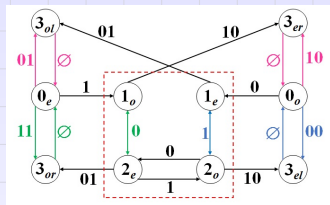
$$\mathcal{E} = \varepsilon(\Pi(\tilde{\mathcal{E}}_l)) = [(\mathcal{D}/3 \setminus \mathcal{D}) \cap (0, 1)] \setminus \varepsilon(\Pi(\mathcal{F})).$$

$$\mathcal{F} = \{\omega \in \Omega_\infty : \omega = u(2_e 2_o)^\infty \text{ for some } u \in \Omega_*\}.$$

Luck (1989) conjecture, Bellissard, Bovier and Ghez (prove) that all the possible gaps of σ_λ are labeled by dyadic numbers or dyadic numbers divided by 3 in $(0, 1)$.

With the coding map in hand, we determine the exact set of numbers which label the gaps.

Lower bound of Hausdorff dimension



$$\tilde{A} := \{1_e, 1_o, 2_e, 2_o\}.$$

$$\tilde{\mathcal{B}}_n := \{I_{0_e w} \in \mathcal{B}_n : w \in \tilde{A}^n\}.$$

$$\text{SNS: } 3_{ol} < 1_o < 3_{or} < 2_o,$$

$$2_e < 3_{el} < 1_e < 3_{er}, 3_{el} < 0_e.$$

$$E := \bigcap_{n \geq 1} \bigcup_{B \in \tilde{\mathcal{B}}_n} B.$$

SNS: Separated Nested Structure.

$$\#\tilde{\mathcal{B}}_n = F_n. \quad E \subset \sigma(H). \quad \forall x \in E, k > 0, |h_k(x)| \leq 2.$$

$$h'_{k+1}(x) = (h_{k-1}^2(x) - 2)h'_k(x) + 2h_{k-1}(x)h_k(x)h'_{k-1}(x).$$

$$|h'_{k+1}(x)| \leq 2|h'_k(x)| + 8|h'_{k-1}(x)|. \quad |h'_k(x)| \leq c4^k.$$

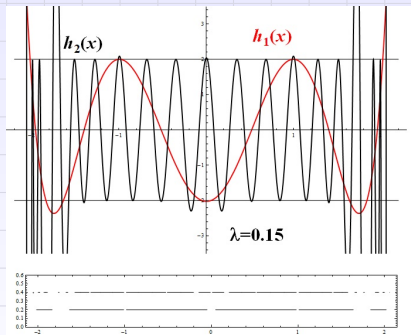
$$B = [x_0, x_1] \in \tilde{\mathcal{B}}_n, \quad 4 = \int_{x_0}^{x_1} |h'_n(x)| dx \leq c(x_1 - x_0)4^n, \quad |B| \geq \hat{c}4^{-n}$$

$$\dim_H(E) \geq \frac{\log \frac{\sqrt{5}+1}{2}}{\log 4}.$$

Generalized Thue-Morse substitution (GTMH)

- Kolar, Ali, Nori, 1991, $\tau(a) = a^m b^n$, $\tau(b) = b^n a^m$

trace map, attractor, electronic spectra, spin excitation, ...



The trace map for $m = n$ is

$$\begin{cases} h_{k+1} = (d_m(h_k))^2(y_k - 2) + 2 \\ y_{k+1} = (d_{2m}(h_k))^2(y_k - 2) + 2, \end{cases}$$

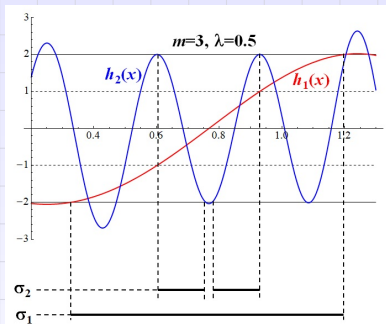
$d_m(x)$: Chebyshev polynomials

$$\sigma_k = \{x : |h_k(x)| \leq 2\}.$$

$$\sigma(H_{m,\lambda}) = \overline{\bigcap_{k \geq 1} \bigcup_{i \geq k} \sigma_i}$$

- We look for **S**eperating **N**ested **S**tructure in the spectrums.

Chebyshev polynomials



$$d_m(2 \cos \theta) = \frac{\sin m\theta}{\sin \theta}.$$

$$d_m(2 \cos \frac{k\pi}{m}) = 0, \quad 1 \leq k < m.$$

$$\text{e.g. } d_3(\pm 1) = 0.$$

$$t \in [-\sqrt{2}, \sqrt{2}], \quad |d'_k(t)| \leq \sqrt{2}k.$$

$$\begin{cases} h_{n+1} = (d_m(h_n))^2(y_n - 2) + 2 \\ y_{n+1} = (d_{2m}(h_n))^2(y_n - 2) + 2 \end{cases}$$

If $d_m(h_n(x)) = 0$, then $h_{n+1}(x) = 2$, $h'_{n+1}(x) = 0$.

$$\dim_H \sigma(H_{m,\lambda}) \geq \frac{\log \max\{2, m-3\}}{\log 64m+4}.$$

Thank you !